

Connectivity of the reconfiguration graph of Hamiltonian paths

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Abstract

A backbite on a Hamiltonian path of a graph G adds an edge of G from an endpoint u of the path to a vertex v and deletes an edge of the path at v so that the result is again a Hamiltonian path. We study the connectivity of the reconfiguration graph $R(G)$, whose vertices are the Hamiltonian paths of G , with two adjacent when a single backbite carries one to the other. We prove that $R(G)$ is connected whenever G contains the square of a Hamiltonian path as a spanning subgraph, and that any two Hamiltonian paths can be reconfigured using $O(n^2)$ backbites. Combining the theorem with known results on spanning squared paths gives connectivity for every graph with $\delta(G) \geq \lceil \frac{2n-1}{3} \rceil$, for the circulants $\text{Circ}(n; S)$ with $a, 2a \in S$ for some a coprime to n , and, with high probability, for the random graph $G(n, p)$ with $p \geq c/\sqrt{n}$, for a sufficiently large constant c . In the other direction, graphs with arbitrarily high connectivity may have disconnected reconfiguration graphs. For every $n \geq 8$ and every $1 \leq m \leq \lfloor n/2 \rfloor - 3$ we construct a graph on n vertices with $\kappa = \lambda = \delta = m$ whose reconfiguration graph is disconnected, with at least $(n - 2m - 4)!$ components. We conjecture that $\delta(G) \geq n/2$ suffices for connectivity.

1 Introduction

A *backbite move* on a Hamiltonian path $w_1 \cdots w_n$ of a graph G chooses an endpoint, say w_1 , and an index $k \geq 3$ with $\{w_1, w_k\} \in E(G)$, adds the edge $\{w_1, w_k\}$, and deletes $\{w_{k-1}, w_k\}$; the result is again a Hamiltonian path. The *reconfiguration graph* $R(G)$ has the Hamiltonian paths of G as its vertices, with two joined when a single backbite carries one to the other, so that distance in $R(G)$ counts backbites. A Hamiltonian path and its reversal are the same vertex of $R(G)$. We study when $R(G)$ is connected, that is, when any two Hamiltonian paths of G can be reconfigured into one another by a sequence of backbites.

The backbite move comes from the Monte Carlo simulation of lattice polymers, where it is used to sample self-avoiding walks [7]. Oberdorf et al. later used the move to sample compact lattice walks and observed that its general ergodicity was not known [11]. The same operation, applied to paths that need not be spanning, is the rotation of Pósa's rotation-extension method [12]. For a survey of reconfiguration problems see [8] and [14]. A *switch* of size k replaces k edges of a Hamiltonian cycle or path of G by k edges of G not on it, so that the result is again a Hamiltonian cycle or path. Takaoka [13] proved that deciding whether two Hamiltonian cycles of a graph are related by a sequence of switches of size two is PSPACE-complete, a question posed implicitly by Ito et al. [5] and raised again in [14]. Kleer, Patel and Stroh [6] proved that any two Hamiltonian cycles of a graph with $\delta(G) \geq n/2 + 7$ are related by switches of size at

most ten, and gave graphs with $\delta(G) \geq 2n/3 - 1$ having two Hamiltonian cycles that no sequence of switches of size two relates. For fixed-endpoint subclasses in rectangular grids, connectivity under switches of size two was proved for simple and 1-complex paths by Nishat, Srinivasan and Whitesides [10, 9]. The author subsequently proved connectivity for all rectangular-grid Hamiltonian paths under a move set consisting of switches of size two, double switches, and backbites [4].

Bousquet et al. [3] reconfigure spanning trees by edge flips, an operation that deletes one edge and adds another. They show that the problem is PSPACE-complete for every bound $d \geq 3$ on the maximum degree of the trees, and they leave the case $d = 2$ open. A spanning tree of maximum degree 2 is a Hamiltonian path, so that case is the problem studied here. A backbite is an edge flip. Conversely, the only flip between Hamiltonian paths that is not itself a backbite joins the two endpoints, and its effect can be realized by rotating the resulting Hamiltonian cycle. Thus edge flips and backbites give the same reachability relation for Hamiltonian paths.

Theorem 1.1 below is the main result. It identifies a structural hypothesis that forces connectivity of $R(G)$. For an ordering $v_1, \dots, v_n$ of a finite set, the *square* of the path $v_1 \cdots v_n$, denoted P_n^2 , is the graph on $\{v_1, \dots, v_n\}$ whose edges join the vertices at index distance 1 or 2.

Theorem 1.1 (Squared-path theorem). Let G be a graph on n vertices containing the square of a Hamiltonian path as a spanning subgraph. The reconfiguration graph $R(G)$ is connected under backbite moves, with diameter $O(n^2)$.

Containing a spanning squared path is preserved under adding edges. If $P_n^2 \subseteq G$ and $G \subseteq G'$ on the same vertex set, then $P_n^2 \subseteq G'$. So any sufficient condition for containing a spanning squared path is itself a sufficient condition for connectivity of $R(G)$, and Corollary 1.2 below records three instances. We first define circulant graphs.

For a set $S \subseteq \mathbb{Z}_n \setminus \{0\}$ with $S = -S$, the *circulant* $\text{Circ}(n; S)$ is the graph with vertex set $\mathbb{Z}_n$ in which i and j are adjacent when $i - j \in S$. We write $\text{Circ}(n; j, k)$ for $\text{Circ}(n; \{\pm j, \pm k\})$, the circulant in which each vertex i is adjacent to $i \pm j$ and $i \pm k$. For a graph F , the *square* F^2 is the graph on $V(F)$ in which two distinct vertices are adjacent whenever their distance in F is at most two.

Corollary 1.2. If G has a Hamiltonian path, then $R(G^2)$ is connected. Moreover, $R(G)$ is connected whenever any of the following hold:

- (a) $\delta(G) \geq \lceil \frac{2n-1}{3} \rceil$;
- (b) $G = \text{Circ}(n; S)$ with $a, 2a \in S$ for some a coprime to n ;
- (c) with high probability, $G = G(n, p)$ with $p \geq c/\sqrt{n}$, for a sufficiently large constant c .

In Section 7 we show that graphs with arbitrarily high connectivity may have a disconnected reconfiguration graph. For every $n \geq 8$ there is a graph on n vertices with minimum degree $\lfloor n/2 \rfloor - 3$ and disconnected reconfiguration graph. Conjecture 7.8 asserts that minimum degree at least $n/2$ suffices.

The proof of Theorem 1.1 is constructive. For a graph G satisfying the hypothesis, we define a canonical Hamiltonian path K of G , and show that any other Hamiltonian path of G can be reconfigured to K by backbites. Since backbites are invertible, it follows that $R(G)$ is connected.

The reconfiguration to K consists of two phases. The first phase reconfigures an arbitrary Hamiltonian path of G into one all of whose edges lie in P_n^2 . This is Corollary 3.4. The second phase reconfigures a Hamiltonian path of P_n^2 into K , and rests on the fact that P_n^2 has exactly one Hamiltonian cycle. This is carried out in Sections 4–5.

The proof of Theorem 1.1 occupies Sections 2–5, and the proof of Corollary 1.2 is given in Section 6. Section 7 constructs graphs whose reconfiguration graph is disconnected.

2 Preliminaries

For a Hamiltonian path H of G , we write $bb(u, v)$ for the backbite of H applied at the endpoint u whose added edge is $\{u, v\}$. The order of the arguments matters precisely when u and v are both endpoints of H , in which case it determines the removed edge: for $H = w_1 \cdots w_n$, the backbite $bb(w_1, w_n)$ removes $\{w_{n-1}, w_n\}$, while the backbite $bb(w_n, w_1)$ removes $\{w_1, w_2\}$. See Figure 1.

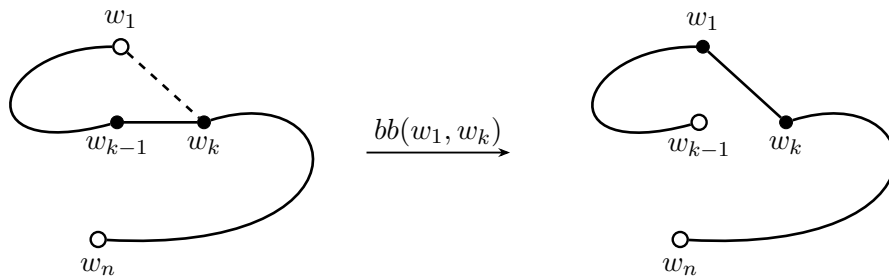


Figure 1: Application of the backbite $bb(w_1, w_k)$ to a path.

For a finite sequence or path $S = (x_1, \dots, x_k)$, S^R denotes its reversal $(x_k, \dots, x_1)$. When one endpoint b of the current path is held fixed over a sequence of backbites, we call the opposite endpoint the *free endpoint*.

Observation 2.1 (Effect of a backbite). Let $H = w_1 \cdots w_n$ be a Hamiltonian path of G and let $bb(w_1, w_k)$, $k \geq 3$, be a backbite of H . The resulting path is $w_{k-1}, \dots, w_1, w_k, \dots, w_n$, and its new endpoint is w_{k-1} . A backbite at w_n is symmetric.

We remark that, by Observation 2.1, a backbite at one endpoint never moves the other endpoint: $bb(w_1, w_n)$ on $w_1 \cdots w_n$ gives $w_{n-1} \cdots w_1 w_n$. Furthermore a backbite is invertible, since after $bb(w_1, w_k)$, we may apply $bb(w_{k-1}, w_k)$ and recover H .

For a Hamiltonian cycle C of a graph and an edge $f = \{x, y\}$ of C , the path $C - f$ has endpoints x and y , and $bb(x, y)$ adds f and removes the edge of C at y other than f , so its result is $C - f'$ for the cycle edge f' neighbouring f at y . Successive such backbites, called *rotations* of C , move the missing edge around C in either direction.

3 Advancing the endpoint over the squared path

Let $P_n = v_1 \cdots v_n$ be a path on n vertices, fixed for the rest of the paper, and let P_n^2 be its square. Throughout this section J is a simple graph on $\{v_1, \dots, v_n\}$ with $E(P_n^2) \subseteq E(J)$. For a Hamiltonian path H of J , $\beta(H) := |E(H) \setminus E(P_n^2)|$ counts its *bad* edges. A backbite whose added edge lies in P_n^2 never increases β , and decreases β exactly when the removed edge is bad.

Fix one endpoint of a Hamiltonian path. Lemma 3.1 shows that backbites at the free endpoint increase its index along the designated ordering $v_1, \dots, v_n$, and repeating this brings the free endpoint to the pair $\{v_{n-1}, v_n\}$. Fixing that endpoint and repeating from the other end

brings both endpoints to $\{v_{n-1}, v_n\}$, and the path can then be closed into a Hamiltonian cycle by adding the edge $\{v_{n-1}, v_n\}$, an edge of P_n^2 . At each step the alternative outcome is a drop in β . Section 5 applies this to G and to P_n^2 .

Lemma 3.1 (Endpoint advancing, one step). Let $n \geq 3$ and let H be a Hamiltonian path of J . Fix one endpoint b of H , and suppose the free endpoint of H is v_i with $i \leq n - 2$. Then there is a sequence of at most two backbites applied at the free endpoint, each with added edge in $E(P_n^2)$, that yields a path H' satisfying one of:

- (a) $\beta(H') < \beta(H)$; or
- (b) the free endpoint of H' is $v_{i'}$ for some $i' > i$.

Proof. Write $H = (v_i, q_2, q_3, \dots)$ from the free endpoint. Since $i \leq n - 2$, both edges $v_i v_{i+1}$ and $v_i v_{i+2}$ lie in $E(P_n^2)$, and only the second vertex q_2 of H is ineligible as the target of a backbite at v_i , so a backbite at v_i with added edge in $E(P_n^2)$ is available. The proof splits into two cases: $q_2 \neq v_{i+2}$ and $q_2 = v_{i+2}$. In each case the required backbite sequences are given explicitly, and each backbite is applied at the current free endpoint with the added edge in $E(P_n^2)$, so β never increases. Thus if the removed edge of some applied backbite is bad, β decreases there, so (a) holds, and we are done. Therefore we need only consider the case where every removed edge lies in $E(P_n^2)$.

CASE 1. $v_{i+2} \neq q_2$. Apply $bb(v_i, v_{i+2})$ to obtain H' with free endpoint v' . By the definition of P_n^2 , $N_{P_n^2}(v_{i+2}) \subseteq \{v_i, v_{i+1}, v_{i+3}, v_{i+4}\}$, so $v' \in \{v_i, v_{i+1}, v_{i+3}, v_{i+4}\}$. Note that by the definition of a backbite, $v' \neq v_i$, and every other index exceeds i , so (b) holds after one backbite. Note that, by the remark after Observation 2.1, the fixed endpoint b remains an endpoint even when the target of the backbite is b itself, as can happen here with $b = v_{i+2}$. This holds for all backbites and will be used implicitly throughout.

CASE 2. $v_{i+2} = q_2$. Apply $bb(v_i, v_{i+1})$ to obtain a path H'' with free endpoint v'' . By the definition of P_n^2 , $N_{P_n^2}(v_{i+1}) \subseteq \{v_{i-1}, v_i, v_{i+2}, v_{i+3}\}$, so $v'' \in \{v_{i-1}, v_i, v_{i+2}, v_{i+3}\}$, and $v'' \neq v_i$ by the definition of a backbite. If $v'' \in \{v_{i+2}, v_{i+3}\}$, then (b) holds after one backbite, with $H'' = H'$.

The only remaining possibility is $v'' = v_{i-1}$, which requires $i \geq 2$: for $i = 1$ there is no vertex v_{i-1} , and the previous sentence already applies. So suppose $i \geq 2$ and $v'' = v_{i-1}$. The removed edge was $\{v_{i-1}, v_{i+1}\}$, so, for some possibly empty subpaths A and C of H ,

$$H = (v_i, v_{i+2}, A, v_{i-1}, v_{i+1}, C), \quad H'' = (v_{i-1}, A^R, v_{i+2}, v_i, v_{i+1}, C).$$

Since $\{v_{i-1}, v_i\} \notin E(H'')$, we may apply $bb(v_{i-1}, v_i)$, obtaining $H' = (v_{i+2}, A, v_{i-1}, v_i, v_{i+1}, C)$ with free endpoint v_{i+2} , so (b) holds after two backbites. □

Corollary 3.2 (Endpoint advancing, iterated). Let H be a Hamiltonian path of J and fix one endpoint b of H . Using backbites at the free endpoint, each with added edge in P_n^2 , one can reach either a path with smaller β , or a path whose endpoints are b and a vertex of $\{v_{n-1}, v_n\}$.

Proof. Apply Lemma 3.1 repeatedly while the free endpoint is v_i with $i \leq n - 2$. If outcome (a) ever occurs, we are done. Otherwise each application increases the index of the free endpoint, which therefore reaches $\{v_{n-1}, v_n\}$ after at most $n - 2$ applications, and the endpoints are then b and a vertex of $\{v_{n-1}, v_n\}$. □

Corollary 3.3 (Smaller β or adjacent endpoints). From every Hamiltonian path of J , backbites with added edges in $E(P_n^2)$ reach a path with smaller β or a path whose endpoints are v_{n-1} and v_n , which are adjacent in P_n^2 .

Proof. Fix either endpoint as b and apply Corollary 3.2. Unless β drops, we reach a path with endpoints b and $w \in \{v_{n-1}, v_n\}$. Now fix w and apply Corollary 3.2 at the opposite end. Unless β drops, the endpoints become w and the vertex of $\{v_{n-1}, v_n\} \setminus \{w\}$, so they are v_{n-1} and v_n , adjacent in P_n^2 . $\square$

Corollary 3.4 (Reaching P_n^2). Backbites reconfigure every Hamiltonian path of J into a Hamiltonian path of P_n^2 .

Proof. Let H be a Hamiltonian path of J . A path with $\beta = 0$ is a Hamiltonian path of P_n^2 , so it suffices to show that from every path with $\beta > 0$ backbites reach a path with smaller β . Apply Corollary 3.3 and let H' be the path reached. If $\beta(H') < \beta(H)$, we are done. Otherwise the endpoints of H' are v_{n-1} and v_n and $\beta(H') = \beta(H) > 0$. Since $v_{n-1}v_n \in E(P_n^2) \subseteq E(J)$, adding $v_{n-1}v_n$ to H' yields a Hamiltonian cycle C of J , and some edge f of C does not lie in $E(P_n^2)$. Rotations of C reconfigure H' into $C - f$, whose edges are those of H' , plus $v_{n-1}v_n$, minus f , so that $\beta(C - f) = \beta(H') - 1$. $\square$

4 The canonical path

Throughout Sections 4–5, G is a simple graph on n vertices, and $v_1, \dots, v_n$ is an ordering of $V(G)$ with $E(P_n^2) \subseteq E(G)$. Such an ordering exists by the hypothesis of Theorem 1.1, and we call it a P_n -ordering of G . Since $E(P_n) \subseteq E(P_n^2)$, the fixed path $P_n = v_1 \cdots v_n$ is a Hamiltonian path of G .

Definition 4.1 (Canonical path). Let G be a simple graph on n vertices and let $v_1, \dots, v_n$ be a P_n -ordering of G . Define the ordering K of $V(G)$ by

$$\begin{aligned} K &= (v_1, v_3, \dots, v_{n-1}, v_n, v_{n-2}, \dots, v_4, v_2) && \text{if } n \text{ is even,} \\ K &= (v_1, v_3, \dots, v_n, v_{n-1}, v_{n-3}, \dots, v_4, v_2) && \text{if } n \text{ is odd.} \end{aligned}$$

Figure 2 illustrates the canonical path for $n = 9$.

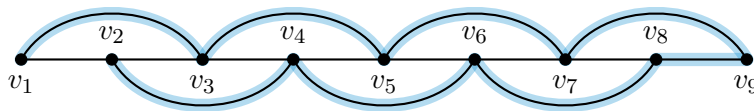


Figure 2: The squared path P_9^2 , with the canonical path $K = v_1v_3v_5v_7v_9v_8v_6v_4v_2$ highlighted.

For the remainder of the paper, K denotes this ordering and no other. It is the canonical form that every Hamiltonian path of G will be reconfigured into.

Lemma 4.2 (The Hamiltonian cycle of P_n^2). Let $n \geq 3$. Then K is a Hamiltonian path of P_n^2 , and $K + \{v_1, v_2\}$ is the only Hamiltonian cycle of P_n^2 .

Proof. Note that consecutive vertices of K are adjacent in P_n^2 , so K is a Hamiltonian path of P_n^2 . Its endpoints v_1 and v_2 are adjacent in P_n^2 , so $K + \{v_1, v_2\}$ is a Hamiltonian cycle of P_n^2 .

It remains to prove uniqueness. Let C be a Hamiltonian cycle of P_n^2 . We will show that $C = K + \{v_1, v_2\}$. For $n = 3$ the graph P_3^2 is a triangle, so this holds. Assume $n \geq 4$. The only neighbours of v_1 are v_2 and v_3 , so C contains the path $v_2v_1v_3$. Suppose that, for some $3 \leq k < n$, the edges of C found so far form a path on $\{v_1, \dots, v_k\}$ with endpoints v_{k-1} and v_k . Each of $v_1, \dots, v_{k-2}$ already has two edges of C , so the second neighbour of v_{k-1} on C is v_k or v_{k+1} . It is not v_k , since that would close a cycle on $k < n$ vertices. So C contains $v_{k-1}v_{k+1}$, and the edges found so far form a path on $\{v_1, \dots, v_{k+1}\}$ with endpoints v_k and v_{k+1} . For $k = n - 1$ this path is spanning, with endpoints v_{n-1} and v_n , and C contains $v_{n-1}v_n$. So C consists of v_1v_2 , the edges v_iv_{i+2} for $1 \leq i \leq n - 2$, and $v_{n-1}v_n$, which are the edges of $K + \{v_1, v_2\}$. $\square$

5 Proof of Theorem 1.1

For $n \leq 3$ the statement is immediate, so we may assume $n \geq 4$. Phase one of the proof takes an arbitrary Hamiltonian path of G to a Hamiltonian path of P_n^2 , by a single application of Corollary 3.4. Phase two is the following proposition.

Proposition 5.1 (Reaching K). Backbites reconfigure every Hamiltonian path of P_n^2 into K .

Proof. Let H be a Hamiltonian path of P_n^2 . Apply Corollary 3.3 with $J = P_n^2$. Its backbites add edges of P_n^2 , so every path they produce is a Hamiltonian path of P_n^2 , and no removed edge is bad. Thus the smaller- β branch cannot occur, and Corollary 3.3 yields a sequence of backbites after which the current path is a Hamiltonian path H' of P_n^2 with endpoints v_{n-1} and v_n . Adding the edge $\{v_{n-1}, v_n\}$ to H' yields a Hamiltonian cycle of P_n^2 , which by Lemma 4.2 is $C = K + \{v_1, v_2\}$. So $H' = C - \{v_{n-1}, v_n\}$. Rotations of C move the missing edge to $\{v_1, v_2\}$, and $C - \{v_1, v_2\} = K$. $\square$

Proof of Theorem 1.1. By Corollary 3.4, applied with $J = G$, backbites reconfigure every Hamiltonian path of G into a Hamiltonian path of P_n^2 . By Proposition 5.1, further backbites reach K . Every Hamiltonian path of G thus reaches K , and $R(G)$ is connected.

We record the bound on the number of backbites. Reaching a Hamiltonian path of P_n^2 costs $O(n^2)$ backbites: each of at most $n - 1$ decreases of β uses one application of Corollary 3.3, itself $O(n)$ backbites, followed by a closing and rotation pass of $O(n)$ backbites. Proposition 5.1 costs $O(n)$ backbites: one application of Corollary 3.3 and at most n rotations. So any Hamiltonian path of G reaches K in $O(n^2)$ backbites, and, backbites being invertible, any two Hamiltonian paths of G are reconfigured into one another by $O(n^2)$ backbites. $\square$

The $O(n^2)$ bound of Theorem 1.1 counts backbites crudely; determining the exact worst-case number is left open. For P_n^2 itself, Proposition 5.1 gives $\text{diam } R(P_n^2) = O(n)$.

6 Proof of Corollary 1.2

For a cyclic ordering $v_1, \dots, v_n$ of a finite set, the *square* of the cycle $v_1 \cdots v_n v_1$, denoted C_n^2 , is the graph on $\{v_1, \dots, v_n\}$ whose edges join the vertices whose indices differ by 1 or 2 modulo n .

Proof of Corollary 1.2. If $v_1 \cdots v_n$ is a Hamiltonian path of F , then $v_iv_{i+1}, v_iv_{i+2} \in E(F^2)$, so that ordering is a P_n -ordering of F^2 , and Theorem 1.1 applies. For the three listed conditions:

- (a) By Fan–Kierstead [1], every graph with $\delta(G) \geq \lceil \frac{2n-1}{3} \rceil$ contains a spanning squared Hamiltonian path; apply Theorem 1.1.
- (b) With $\gcd(a, n) = 1$, the ordering $0, a, 2a, \dots, (n-1)a$ visits all of $\{0, \dots, n-1\}$; consecutive terms differ by $a \in S$ and terms two apart by $2a \in S$, so this ordering spans $C_n^2 \supseteq P_n^2$; apply Theorem 1.1.
- (c) By Kahn–Narayanan–Park [2], there is a constant $c > 0$ such that, if $p \geq c/\sqrt{n}$, then the random graph $G(n, p)$ with high probability contains the square of a Hamilton cycle, hence a spanning copy of P_n^2 ; Theorem 1.1 applies to every graph containing one.

□

7 Graphs with a disconnected reconfiguration graph

It is natural to ask which hypotheses weaker than that of Theorem 1.1 force $R(G)$ to be connected. Edge density alone is not enough. Fix $t \geq 2$ and consider the graph H_t obtained from the complete graph on the $t+2$ vertices $a, b, h_1, \dots, h_t$ by deleting the edge ab and adding two vertices x and y , with x joined to a and y joined to b . Every Hamiltonian path of H_t runs from x to y and admits no backbite, so $R(H_t)$ has no edges. See Figure 3. Since $t \geq 2$, there is more than one Hamiltonian path, and $R(H_t)$ is disconnected. Yet H_t has only $2t+4$ fewer edges than the complete graph on $t+4$ vertices, so its edge density tends to one. The two vertices of degree one force this, but they are not needed for disconnection. A small grid graph with disconnected $R(G)$, the union of two 4×4 squares offset by one unit, is given in [4]. The construction below reaches, at every order, every connectivity and minimum degree up to $\lfloor n/2 \rfloor - 3$.

Theorem 7.1 (Disconnection at every connectivity). For every $n \geq 8$ and every $1 \leq m \leq \lfloor n/2 \rfloor - 3$ there is a graph G on n vertices with $\kappa(G) = \lambda(G) = \delta(G) = m$ whose reconfiguration graph $R(G)$ is disconnected, with at least $(n - 2m - 4)!$ components.

The graph G of Theorem 7.1 is $A_m(H_t)$, where $A_m(H)$ is defined as follows. For a connected graph H and $m \geq 1$, let $S = \{s_1, \dots, s_m\}$ and $Z = \{z_1, \dots, z_m\}$ be disjoint sets of new vertices. The graph $A_m(H)$ is the complete bipartite graph with parts S and $V(H) \cup Z$, together with the edges of H . That is,

$$V(A_m(H)) = V(H) \cup S \cup Z, \quad E(A_m(H)) = E(H) \cup \{uv : u \in S, v \in V(H) \cup Z\}.$$

So S and Z are independent sets, the vertices of Z have all their neighbours in S , and $A_m(H) - S$ is the disjoint union of H and the m isolated vertices $z_1, \dots, z_m$. The order of $A_m(H)$ is $|V(H)| + 2m$. Figure 3 shows $A_2(H_2)$.

Observation 7.2 (Path structure). Let H be a connected graph, let $m \geq 1$, and let P be a Hamiltonian path of $A_m(H)$. Then $P - S$ has exactly $m+1$ subpaths, namely a Hamiltonian path of H and the m single-vertex paths $z_1, \dots, z_m$. Moreover no vertex of S is an endpoint of P .

Proof. Removing m vertices from a path leaves at most $m+1$ subpaths. Each subpath of $P - S$ is connected in $A_m(H) - S$ and so lies in a single component. P is Hamiltonian, so each of the $m+1$ components of $A_m(H) - S$ contains a subpath of $P - S$. Hence $P - S$ has exactly $m+1$ subpaths, one in each component. The subpath in the component H spans $V(H)$, and its edges lie in $E(H)$, so it is a Hamiltonian path of H . The subpath in $\{z_i\}$ is the single vertex z_i .

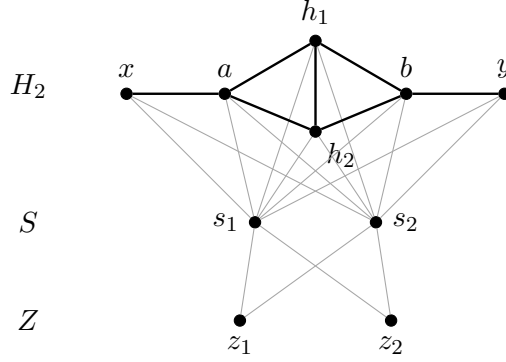


Figure 3: The graph $A_2(H_2)$, of order 10. The edges of H_2 are black and the edges from S are grey. Deleting S leaves H_2 and the two isolated vertices z_1 and z_2 .

Suppose some vertex of S is an endpoint of P . Deleting it does not split the path, and deleting each of the remaining $m - 1$ vertices of S increases the number of subpaths by at most one, so $P - S$ has at most m subpaths, a contradiction. So no vertex of S is an endpoint of P . $\square$

We write P_H for the Hamiltonian path of H contained in P , given by Observation 7.2.

Observation 7.3 (Hamiltonian paths of H_t). Let $t \geq 2$. Then H_t has exactly $t!$ Hamiltonian paths, and every one runs from x to y , with second vertex a and second-to-last vertex b .

Lemma 7.4 (The subpath on H_t is preserved). Let $t \geq 2$, let $m \geq 1$, and let P be a Hamiltonian path of $A_m(H_t)$. Then every backbite of P leaves $E(P_{H_t})$ unchanged.

Proof. Apply a backbite at the endpoint u of P , with target v . If v lies in S , then neither the added edge nor the removed one has both ends in $V(H_t)$, and so $bb(u, v)$ leaves $E(P_{H_t})$ unchanged. Thus it is enough to show that v lies in S . By Observation 7.2, u is not on a vertex of S , so u lies in Z or u is an endpoint of P_{H_t} . If u lies in Z , then all its neighbours lie in S , so $v \in S$. Otherwise, by Observation 7.3, u is on x or y . Suppose u is on x . The neighbour of x in P is its neighbour a in P_{H_t} , and $N(x) = \{a\} \cup S$. It follows that $v \neq a$, and so v must lie in S . A similar argument works if u is on y . $\square$

Corollary 7.5 (Component count). Let $t \geq 2$ and $m \geq 1$. Then $R(A_m(H_t))$ has at least $t!$ components.

Proof. By Lemma 7.4, $E(P_{H_t})$ remains unchanged under backbites, so it is the same for all Hamiltonian paths in a component of $R(A_m(H_t))$. Every Hamiltonian path Q of H_t extends to the Hamiltonian path $(Q, s_1, z_1, s_2, z_2, \dots, s_m, z_m)$ of $A_m(H_t)$, whose subpath on $V(H_t)$ is Q . Distinct paths have distinct edge sets, and H_t has exactly $t!$ Hamiltonian paths by Observation 7.3, so there are $t!$ possibilities for $E(P_{H_t})$, and each of them occurs. Hence $R(A_m(H_t))$ has at least $t!$ components. $\square$

Proposition 7.6 (Connectivity and degree of $A_m(H)$). Let H be a connected graph and let $m \geq 1$. Then $\kappa(A_m(H)) = \lambda(A_m(H)) = \delta(A_m(H)) = m$.

Proof. Deleting the edges of H from $A_m(H)$ leaves the complete bipartite graph with parts S and $V(H) \cup Z$, for which $\kappa = \lambda = \delta = m$. Adding edges decreases none of the three, so $\kappa(A_m(H)) \geq m$. Every vertex of Z has neighbourhood S , so $\delta(A_m(H)) = m$. Since $\kappa \leq \lambda \leq \delta$, all three equal m . $\square$

Proof of Theorem 7.1. Let $n \geq 8$ and $1 \leq m \leq \lfloor n/2 \rfloor - 3$, and set $t = n - 2m - 4$, so that $t \geq 2$. Let $G = A_m(H_t)$, of order $t + 4 + 2m = n$. By Proposition 7.6, $\kappa(G) = \lambda(G) = \delta(G) = m$. By Corollary 7.5, $R(G)$ has at least $t! = (n - 2m - 4)!$ components, and $t! \geq 2$, so $R(G)$ is disconnected. $\square$

Remark 7.7. Every backbite of $A_m(H)$ either leaves $E(P_H)$ unchanged or induces a single backbite on P_H , and every Hamiltonian path of H extends to one of $A_m(H)$ as in the proof of Corollary 7.5. So for any connected graph H , $R(A_m(H))$ is disconnected whenever $R(H)$ is.

Fixing $j \geq 2$ and letting m grow, the graphs $A_m(H_j)$ have $\delta/n \rightarrow \frac{1}{2}$ and at least $j!$ components.

Conjecture 7.8. Let G be a graph on n vertices with minimum degree at least $n/2$. Then the backbite reconfiguration graph $R(G)$ is connected, that is, any two Hamiltonian paths of G can be reconfigured into one another by a sequence of backbite moves.

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